

Wētā FX Presents: Avatar: Fire and Ash

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Figure 1: Wētā FX Presents: Avatar: Fire and Ash

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1 Overview

In *Avatar: Fire and Ash*, we return to Pandora's environments and biomes, creatures and clans, and immerse audiences in water, and dazzle them with fire. The Academy Award winning VFX team from Wētā FX and Lightstorm Entertainment will overview the process of creating the film's 3,100+ shots including how the teams

blended extensive native stereo photography with CG environments and fully digital performance-captured characters. While advanced tools are required to work at such scale, everything was handcrafted - every character, tree, leaf, pore, sweat drop and insect - to ensure Pandora is a believable world. To open the session, Nicky, and Peter, will outline the logistical challenges Fire and Ash presented: managing, scheduling, and supporting a team of four Visual Effects Supervisors, six Sequence VFX Supervisors, two Animation Supervisors, and more than 1,200 crew over the course of roughly three years. sharing learnings of how 1 and 2 helped chart a smoother path. Eric will discuss how the film is the product of 20 years of close collaboration and technical achievements. Wētā FX and Lightstorm entertainment, have worked on Avatar projects for the last 20 years, establishing a creative shorthand and shared vocabulary that have enabled the highest quality of storytelling and visual spectacle. Eric will also discuss how Wētā FX's philosophy for Fire and Ash was the same as on other past projects: anchoring the visuals in real world science, before infusing human artistry and technical innovations to bring the world of Pandora to life. Richie will review how the virtual cinematography in the film—and how the film was shot natively in stereo 3D—meant that every shot had to be accurate not just on a flat plane, but also in 3D space as if it

was a volume. He will review the shot composition process, and how performance capture, motion and animation is all considered. Pavani will take a deep dive into the creation and destruction of the Windtrader fleet. She will discuss how we built the complex models and dealt with the challenges of scale, movement, rendering and simulations, and creative strategies used to realize the director's vision. Nic, dives into some of the new effects challenges in the film—more water, bigger models, and of course, how the team tackled the titular element, fire, created in Kora, a new addition to Wētā FX's global simulation framework Loki, allowing artists to creatively direct repeatable combustion physics while maintaining physical plausibility. Finally, Dan will discuss the film's characters and creatures, highlighting how the team of facial animators took great care to preserve each actor's performance and creative intent, using technology and human artistry to breathe life into the emotional heart of the film. Wētā FX's team created over 2,500 facial motion shots for 30 speaking characters, including Oona Chaplin's Varang, using a proprietary facial system that uses a blend of neural networks to solve muscles embedded in the face volume and allows layers of animator adjustments in a physically plausible manner, preserving the artistic performance of the actor.